

Suggested Solutions to Exercises 6

1. (a) From the properties of BM, the mean at time T is zero, and the variance is T .
- (b) If the process is fixed at k at $t = 1$, we need to consider the increment between time $t = 1$ and time $t = 3$. From the properties of BM, the distribution of the increment $W_3 - W_1$ will be normal, with mean zero and variance equal to the length of the time period, which is $3 - 1 = 2$. Thus, conditional on $W_1 = k$, the mean of W_3 is $k + 0 = k$, and the variance is 2.
2. We have that

$$S_6 = (S_6 - S_3) + (S_3 - S_0) + S_0$$

By the properties of the generalised BM we know that

$$\begin{aligned}(S_3 - S_0) &\sim N(2 \times 3, 3^2 \times 3) \\ (S_6 - S_3) &\sim N(3 \times 3, 4^2 \times 3)\end{aligned}$$

Also, the two above increments are independent. It follows that:

$$S_6 \sim N(2 \times 3 + 3 \times 3 + 5, 3^2 \times 3 + 4^2 \times 3)$$

We conclude that

$$S_6 \sim N(20, 75)$$

3. Let the company's initial cash position be C_0 . Then the distribution of C_1 (i.e., the cash position after 1 year) is

$$C_1 \sim N(C_0 + 4 \times 0.5, 4 \times 4)$$

It follows that

$$P[C_1 < 0] = P\left[\frac{C_1 - C_0 - 2}{4} < \frac{-C_0 - 2}{4}\right] = N\left(\frac{-C_0 - 2}{4}\right)$$

where $N(\cdot)$ is the cdf of the standard normal distribution. From the table of the standard normal distribution we have

$$N\left(\frac{-C_0-2}{4}\right) = 0.05 \quad \Leftrightarrow \quad \frac{-C_0-2}{4} = -1.6499$$

Solving for C_0 we obtain $C_0 = 4.5996$. Thus, the initial cash position must then approximately £4.6 millions.

4. We need to verify for Z_t the properties that define BM. Since W_t is BM, it has continuous paths by definition, and clearly $Z_t = -W_t$ also has continuous paths. We also have that $Z_0 = -W_0 = 0$, thus $\{Z_t\}$ satisfies another requirement of BM.

Consider the increments, for $0 < s < t$,

$$Z_t - Z_s = -W_t + W_s = -(W_t - W_s)$$

We know that $W_t - W_s \sim N(0, t - s)$, so also $Z_t - Z_s \sim N(0, t - s)$. Thus, $\{Z_t\}$ satisfies the BM property of normal increments.

Since $\{W_t\}$ has independent increments, we know that $W_{t_2} - W_{t_1}$ is independent of $W_{s_2} - W_{s_1}$ for $s_1 < s_2 \leq t_1 < t_2$. This means that $-Z_{t_2} + Z_{t_1}$ is independent of $-Z_{s_2} + Z_{s_1}$, thus $Z_{t_2} - Z_{t_1}$ is independent of $Z_{s_2} - Z_{s_1}$. In other words, $\{Z_t\}$ has independent increments.

Therefore $\{Z_t\}$ is a BM, as it possesses all the properties of a BM.

5. The pdf of B_t is equal to

$$f(x; t) = \frac{1}{\sqrt{2\pi t}} \exp\left\{-\frac{x^2}{2t}\right\}.$$

Taking partial derivatives with respect to t and x , we obtain

$$\frac{\partial f}{\partial t} = \frac{x^2 - t}{2t^2} f, \quad \frac{\partial f}{\partial x} = -\frac{x}{t} f, \quad \frac{\partial^2 f}{\partial x^2} = \frac{x^2 - t}{t^2} f = 2 \frac{\partial f}{\partial t},$$

so that the given diffusion equation is satisfied.

6. Without loss of generality we assume that $s < t$ so that $\min\{s, t\} = s$.

- (a) **Long Approach - Not Recommended:** Using the properties of BM and conditioning on the value of the process at time s , we can write the joint pdf of W_t, W_s as

$$f_{W_s, W_t}(x, y) = p(s, 0, x) p(t - s, x, y)$$

where $p(\cdot, \cdot, \cdot)$ is the pdf of a normal distribution $N(\mu, t)$, that is

$$p(t, \mu, x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-\mu)^2}{2t}}$$

Thus, we have

$$\begin{aligned} \mathbb{E}[W_s W_t] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy p(s, 0, x) p(t - s, x, y) dx dy \\ &= \int_{-\infty}^{\infty} x p(s, 0, x) \left(\int_{-\infty}^{\infty} y p(t - s, x, y) dy \right) dx \end{aligned}$$

The inner integral can be written as

$$\begin{aligned} \int_{-\infty}^{\infty} y p(t - s, x, y) dy &= \int_{-\infty}^{\infty} (x + u) p(t - s, x, x + u) du \\ &= \int_{-\infty}^{\infty} (x + u) p(t - s, 0, u) du \\ &= x \int_{-\infty}^{\infty} p(t - s, 0, u) du + \int_{-\infty}^{\infty} u p(t - s, 0, u) du \\ &= x + 0 \end{aligned}$$

where we have used the substitution $u = y - x$ in the first step, and properties of the normal density thereafter. Thus, returning to the expression for $\mathbb{E}[W_s W_t]$, we have

$$\mathbb{E}[W_s W_t] = \int_{-\infty}^{\infty} x^2 p(s, 0, x) dx = s$$

where we have again used the properties of the pdf of the normal distribution, and have corresponded the integrals to simple moments of an appropriately parameterized normal distribution. As we assumed that $\min\{s, t\} = s$, we have indeed obtained

$$\mathbb{E}[W_s W_t] = \min\{s, t\}$$

as required.

- (b) **Fast Approach - Highly Recommended:** We use iterative conditional expectation, and obtain that (again, we have assumed that $s < t$)

$$\begin{aligned} \mathbb{E}[W_s W_t] &= \mathbb{E}[\mathbb{E}[W_s W_t | W_s]] \\ &= \mathbb{E}[W_s \mathbb{E}[W_t | W_s]] \\ &= \mathbb{E}[W_s W_s] = s \end{aligned}$$